

Software Requirement Specification (SRS)

E-Commerce Order Management Database System

1. Introduction

1.1 Purpose

This Software Requirement Specification (SRS) document describes the functional and non-functional requirements for the E-Commerce Order Management Database System. It is intended to serve as the foundation for subsequent database design activities, including ER modelling, normalization, and SQL implementation, and is intended for use by the project's database designers, developers, and academic evaluators.

1.2 Scope

The system will provide a centralized database to manage customer registration, product and category management, supplier information, inventory tracking, order processing, payment management, shipment tracking, and customer reviews for a growing e-commerce business. It aims to reduce data redundancy, maintain data accuracy, and enable meaningful business reporting. The scope of the current phase is limited to requirement analysis and specification; user-interface, third-party payment/courier integration, and mobile applications are out of scope, as detailed in the accompanying Business Requirement Document.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirement Specification
ER Model	Entity-Relationship Model, used to visually represent data and its relationships
PK	Primary Key — a unique identifier for a record in a table
FK	Foreign Key — an attribute that references the primary key of another table
DBMS	Database Management System
COD	Cash on Delivery, a payment method

1.4 References

- Week 1 project brief — "Understanding the Business Problem", E-Commerce Order Management Database System semester project.
- Requirement Analysis Report (companion document, this submission).
- Business Requirement Document (companion document, this submission).

1.5 Overview

Section 2 of this document describes the product from a general perspective. Section 3 lists the specific functional and non-functional requirements, along with the data entities that the database must support. Section 4 provides supporting appendices, including the full entity-attribute reference used throughout the project.

2. Overall Description

2.1 Product Perspective

The E-Commerce Order Management Database System is a new, self-contained relational database intended to replace fragmented manual and spreadsheet-based record keeping. It is the data backbone for an online shopping platform and is designed to be consumed by future application layers (web/mobile storefronts, admin consoles, reporting dashboards) that are outside the scope of this phase.

2.2 Product Functions (Summary)

- Register and maintain customer profiles.
- Organize products into categories and maintain supplier information.
- Track product stock levels.
- Process customer orders and their line-item details.
- Record and track payments against orders.
- Record and track shipments and delivery status.
- Capture customer reviews and ratings for products.
- Support generation of business reports from the above data.

2.3 User Classes and Characteristics

User Class	Characteristics
Customer	Registers an account, places orders, makes payments, and submits reviews. Assumed to have basic digital literacy and no database expertise.
Catalogue / Inventory Administrator	Manages products, categories, suppliers, and stock levels. Requires access to product and inventory data.
Order Fulfilment & Logistics Staff	Processes orders, confirms payments, and updates shipment status. Requires access to order, payment, and shipment data.
Business / Management	Consumes reports derived from the database; requires read/aggregate access rather than transactional access.
Database Administrator	Designs, secures, and maintains the database; requires full administrative access.

2.4 Operating Environment

The system is designed to run on a standard relational Database Management System (e.g. MySQL, PostgreSQL, or SQL Server) hosted on a server environment accessible to the application layers described in Section 2.1. No specific hardware constraints are assumed beyond what is typical for an academic project environment.

2.5 Design and Implementation Constraints

- Only the nine predefined core entities — Customer, Category, Product, Supplier, Order, Order Details, Payment, Shipment, and Review — and their listed attributes may be used.

- The database must be designed to support normalization (up to at least 3NF) in later project weeks without requiring changes to the core entity list.
- The project must be implemented and version-controlled using a GitHub repository.

2.6 Assumptions and Dependencies

This SRS assumes that the business scenario, core entities, and their attributes as defined in the Week 1 project brief remain fixed for the duration of the project. Any change to these would require this specification to be revisited. Full details of assumptions and constraints are provided in the Business Requirement Document.

3. Specific Requirements

3.1 Functional Requirements

The complete, itemized functional requirements are provided in Section 2 of the accompanying Business Requirement Document and are summarized by module below.

Module	Summary of Functional Requirements
Customer Management	Register, update, and retrieve customer profile and order history.
Category & Product Management	Create, update, and search products and categories.
Supplier Management	Maintain supplier records linked to supplied products.
Inventory Management	Adjust and validate stock quantity against orders and restocking.
Order & Order Details Management	Place orders, record line items, track order status, compute order totals.
Payment Management	Record payment method, date, and status against an order.
Shipment Management	Record delivery address, shipment date, and delivery status.
Review Management	Capture customer ratings and comments for products.
Reporting	Generate sales, revenue, delivery, and product-rating reports.

3.2 Data Requirements — Entities and Attributes

The database must implement the following entities and attributes exactly as defined, with no additions, removals, or renaming.

Entity	Attributes
Customer	Customer ID (PK), Name, Email, Mobile Number, Address, Registration Date
Category	Category ID (PK), Category Name, Description
Product	Product ID (PK), Product Name, Price, Stock Quantity, Category ID (FK)
Supplier	Supplier ID (PK), Supplier Name, Contact Information
Order	Order ID (PK), Customer ID (FK), Order Date, Total Amount, Order Status
Order Details	Order Detail ID (PK), Order ID (FK), Product ID (FK), Quantity, Unit Price
Payment	Payment ID (PK), Order ID (FK), Payment Method, Payment Date, Payment Status
Shipment	Shipment ID (PK), Order ID (FK), Delivery Address, Shipment Date, Delivery Status
Review	Review ID (PK), Customer ID (FK), Product ID (FK), Rating, Comments

3.3 External Interface Requirements

At this stage the database has no external interfaces of its own; it is expected to expose its data to future application layers (storefront, admin console, reporting tools) via standard SQL queries or an

intermediate API, both of which are out of scope for this phase.

3.4 Non-Functional Requirements

The complete non-functional requirements — performance, scalability, data integrity, security, reliability, usability, maintainability, auditability, and backup/recovery — are detailed in Section 3 of the accompanying Business Requirement Document and apply to this specification in full.

4. Appendix

4.1 Entity Relationship Summary (Narrative)

- A Category classifies many Products (one-to-many).
- A Supplier supplies many Products (one-to-many).
- A Customer places many Orders (one-to-many).
- An Order contains many Order Details lines, and each Order Details line references one Product (resolving the many-to-many relationship between Order and Product).
- An Order has one associated Payment record (one-to-one).
- An Order has one associated Shipment record (one-to-one).
- A Customer writes many Reviews, and each Review refers to one Product (many-to-one from Review to both Customer and Product).

4.2 Document Control

This SRS, together with the Requirement Analysis Report and Business Requirement Document, forms the complete Week 1 deliverable set for the E-Commerce Order Management Database System semester project and will be maintained in the project's GitHub repository alongside the README file.